

Name: _____ Date: _____

Period: _____

1. **Difference table.** Complete the table and identify the polynomial degree.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	2	—	—	—
1	6	_____	—	—
2	16	_____	_____	—
3	38	_____	_____	_____
4	78	_____	_____	_____
5	142	_____	_____	_____

The _____ differences are constant, so this data is best modeled by a degree-_____ polynomial.

2. **Context-to-model matching.** Match each scenario (A–D) to the best model type (1–4).

Scenario

Model type

A. Water drains from a cylindrical tank; volume depends on the cube of the radius. 1. Linear

B. A taxi charges a flat fee plus a fixed rate per mile. 2. Quadratic

C. A ball is launched; height depends on horizontal distance. 3. Cubic

D. Profit per unit is constant regardless of units sold. 4. Piecewise

A → _____ B → _____ C → _____ D → _____

3. **Assumption articulation.** A linear function models the distance d (in miles) a cyclist travels as a function of time t (in hours).

(a) State one assumption this model makes about the cyclist's speed.

(b) Give a domain restriction and explain why it applies.

Domain: _____ Reason: _____

4. **Piecewise model description.** A parking garage charges \$3.00 for the first hour and \$1.50 per hour (or fraction thereof) for each additional hour, up to a daily maximum of \$15.00. Describe a piecewise model for the cost C as a function of hours h .

Piece 1 (when _____): $C =$ _____ Type: _____

Piece 2 (when _____): $C =$ _____ Type: _____

Piece 3 (when _____): $C =$ _____ Type: _____

5. **Model selection with justification.** A scientist measures the spread of a chemical across a flat surface. The affected area (in cm^2) at several distances r (in cm) from the source is recorded below.

r	A	ΔA	$\Delta^2 A$
1	3	—	—
2	12	_____	—
3	27	_____	_____
4	48	_____	_____

Based on differences, what model type fits? _____

Is there a context clue that supports this choice? Explain in one sentence.

Extension optional

Extension (optional). The table below lists four data points. A student claims a cubic polynomial is the unique model because there are 4 points. Is this claim correct? Explain by computing differences and stating the degree of the unique polynomial.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	0	—	—	—
1	2	_____	—	—
2	8	_____	_____	—
3	18	_____	_____	_____